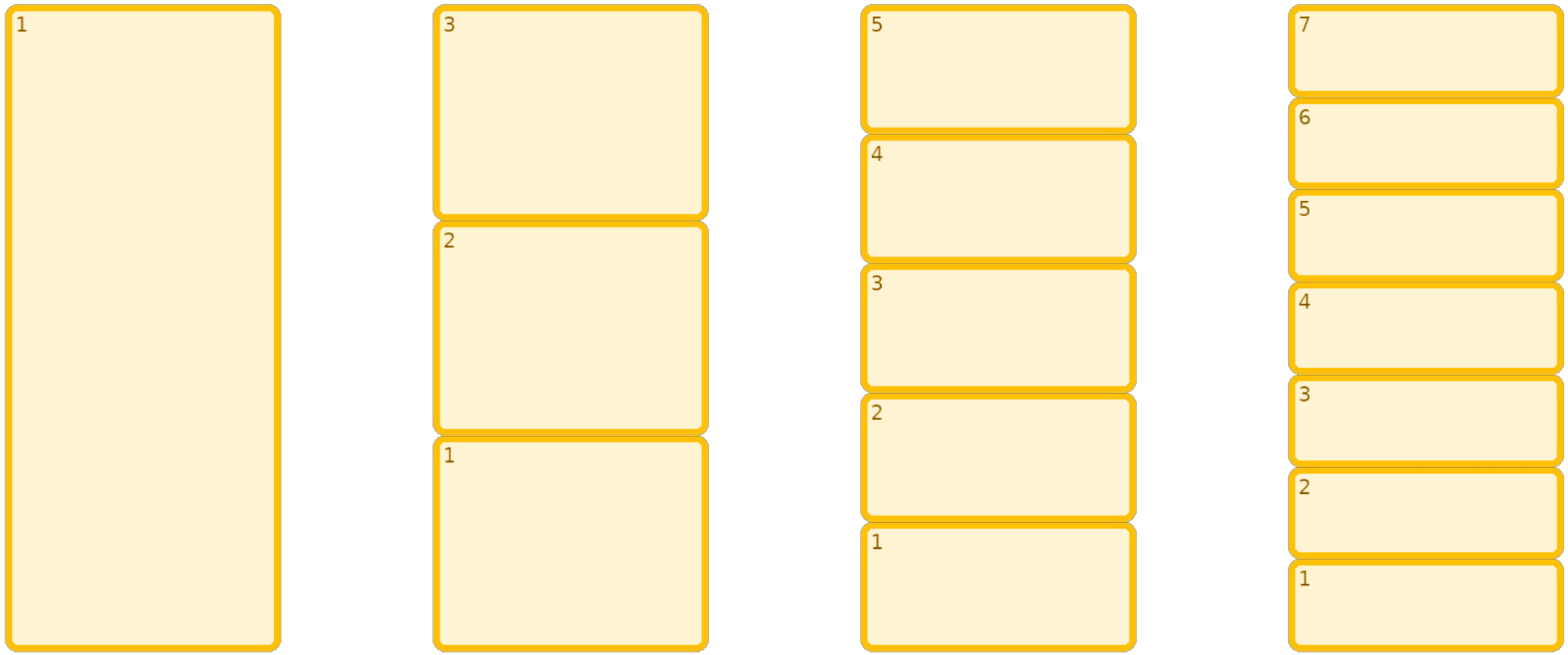
# Ling 285
# Human Language and Technology
# Lecture 14

# Speech → Neural Networks

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1
3
2
1
5
4
3
2
1
7
6
5
4
3
2
1

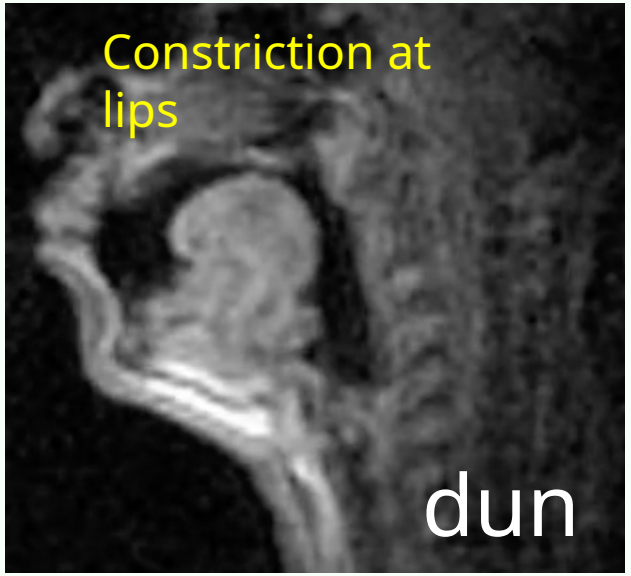
# Pop quiz

A B C D E F

# Vowels

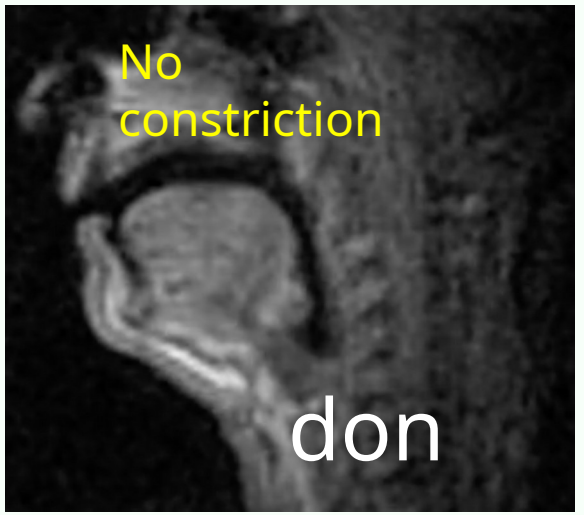
High



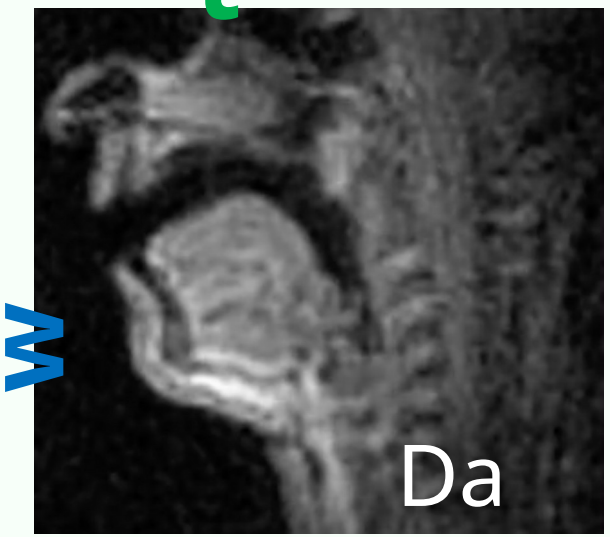

Front

Back

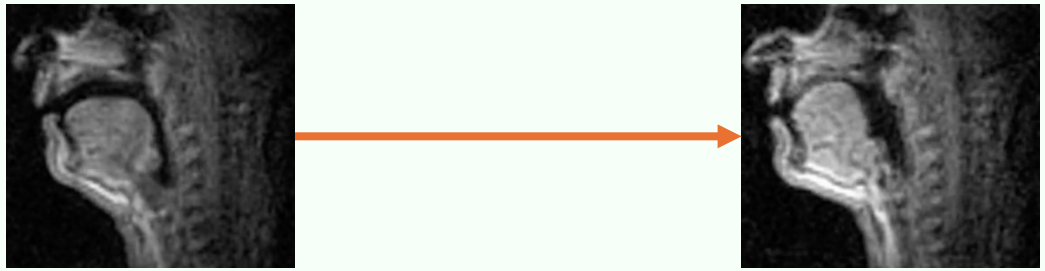



Central

Low





When we compress air, as happens in a constriction, its stiffness rises, so making a constriction at a place that experiences lots of compression, will raise frequency. And bunching up air makes its mas//density larger, so in a part of the vocal tract where air moves a lot, constricting raises mass! So making a constriction increases mass and increases stiffness.

But be careful!!!! This doesn’t mean that constrictions have no effect on frequency!!! Why? Because the parts of a vibrating medium that moves the most alternate with those that compress/uncompress the most!

**/u/ = free-end loading**

**Mode 1: motion dominates**

**Mode 2: motion dominates**

**F1 ↓  F2 ↓**

Click to reveal the reasoning steps.

**/i/ = constriction near 2/3**

**Mode 1: mass motion dominates**

**Mode 2: springiness dominates**

**F1 ↓  F2 ↑**

Click to reveal the reasoning steps.

**/a/ = constriction near 1/3**

**Mode 1: springiness dominates**

**Mode 2: mass motion dominates**

**F1 ↑  F2 ↓**

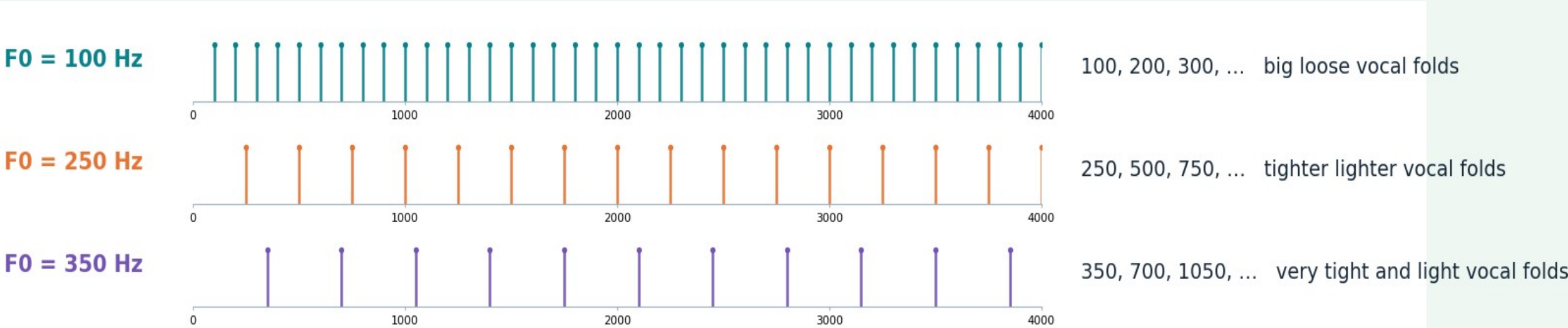
Click to reveal the reasoning steps.

# Source: membrane modes and vocal-fold harmonic spacing

Top: fixed–fixed membrane modes. Bottom: three source spectra before the vocal-tract filter.

## Fixed–fixed: five modes plus their sum




Same 0–4000 Hz window. Lower F0 gives denser harmonic sampling; higher F0 gives sparser sampling.

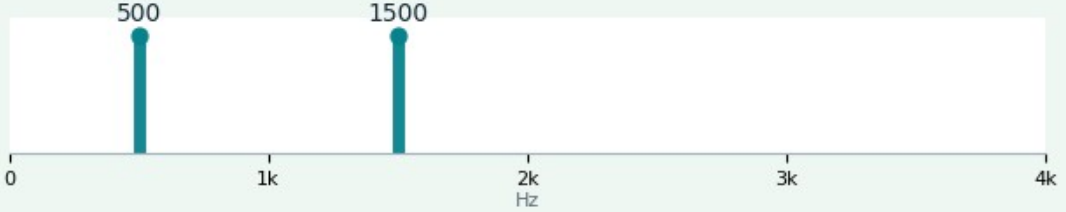
# Filter: the fixed–free vocal tract selects formants

A formant pulse makes nearby harmonics audible and mutes far-away harmonics.

## Fixed–free: one end closed, one end open




**Schwa /ə/**







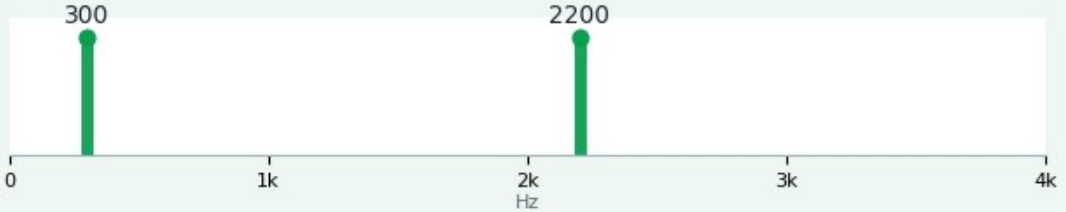
**/u/**





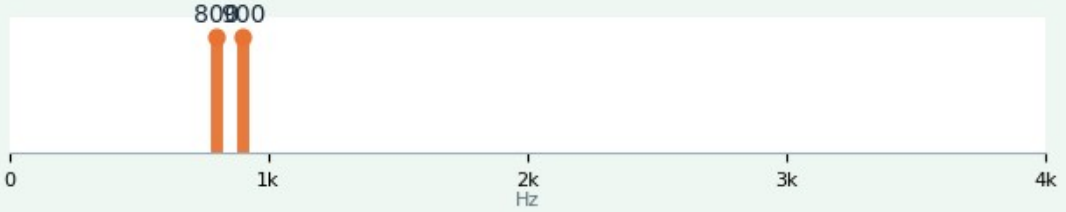


**/i/ or /I/**






**/a/**




## Core rule

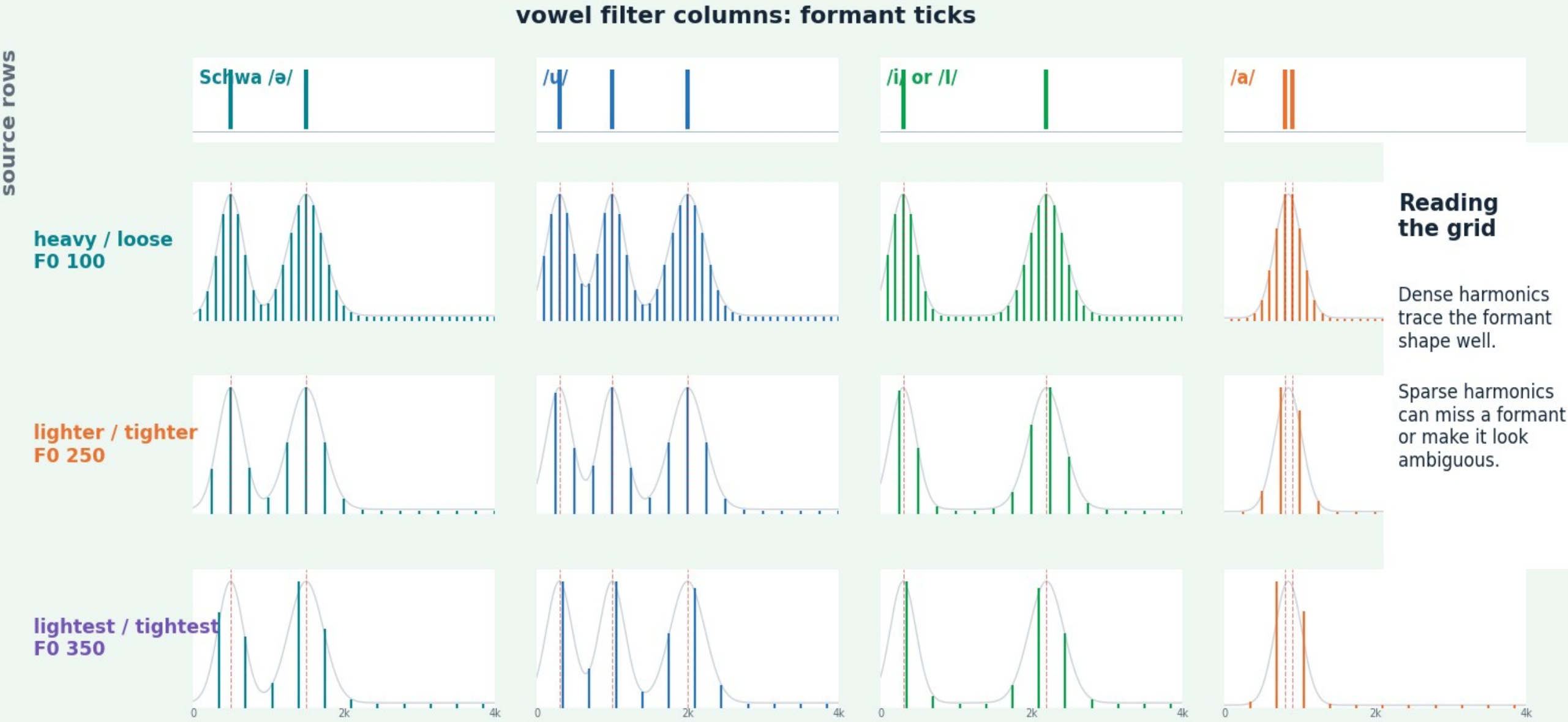
A formant encourages the closest harmonics to be audible.

Harmonics far from that formant are muted.

The vowel shape is the filter. The voiced source supplies harmonics; the filter chooses which harmonics survive.

# Source × filter: same vowel filter, different harmonic sampling

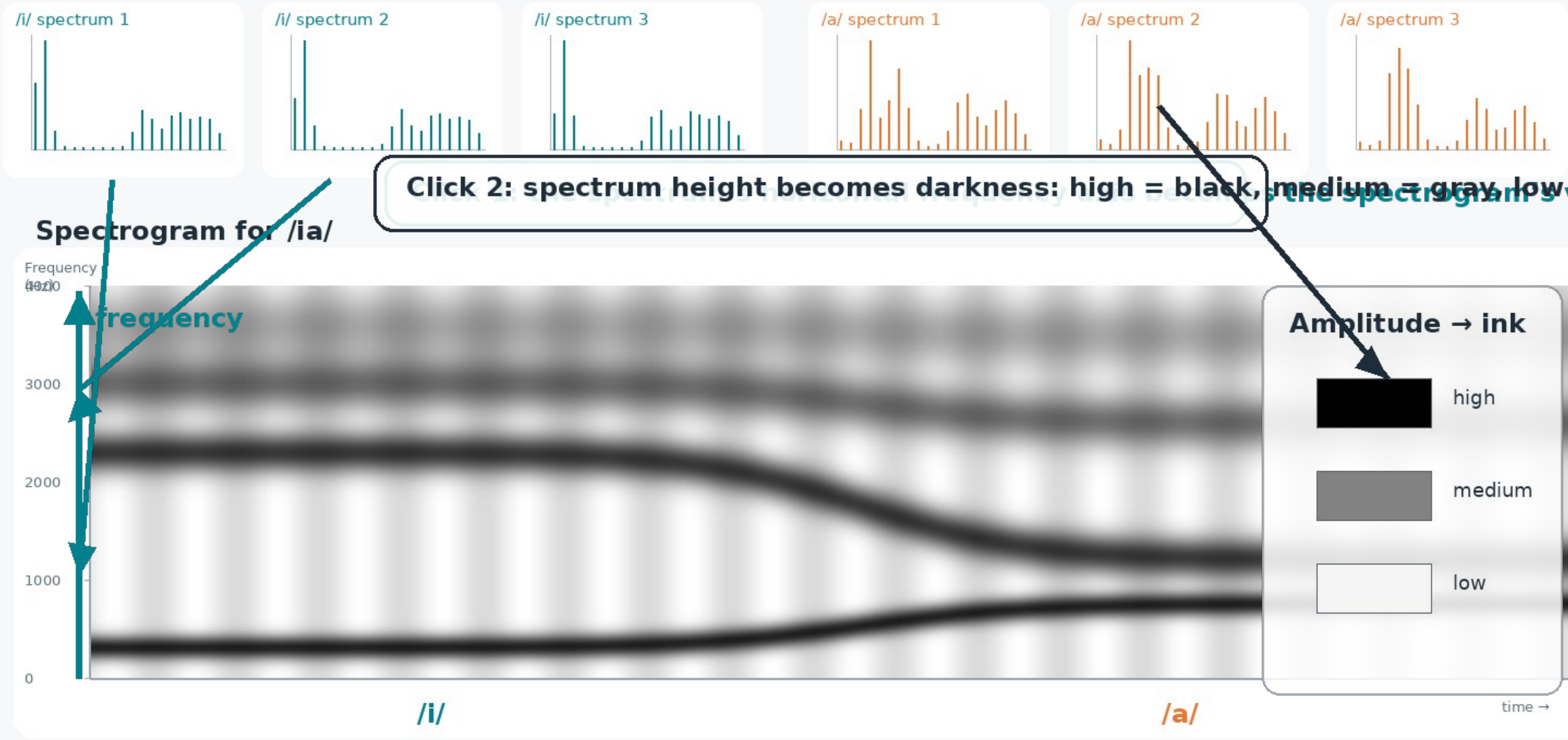
More harmonics in the window give a smoother, easier-to-read shaped spectrum




Dashed red ticks = formant targets. Colored stems = actual harmonics after filtering.

# Spectrum versus spectrogram

A spectrum is one instant. A spectrogram stacks many spectra over time.



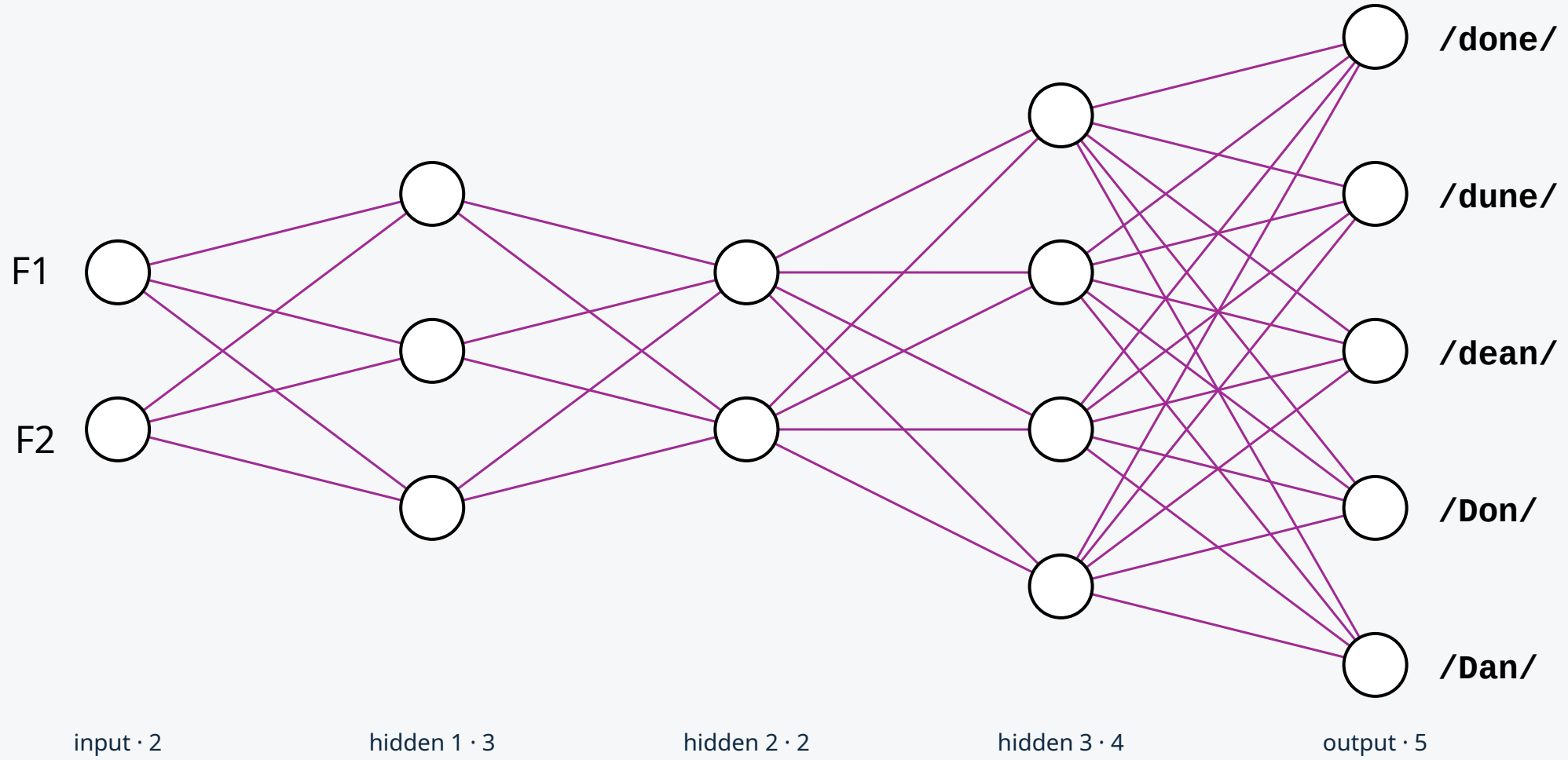
# Where were you a year ago?

# My favorite experiment to show how complex speech is!!!!!

# Back to AI: We will build an NN that will be trained on your Formants and will decide which vowel is being spoken

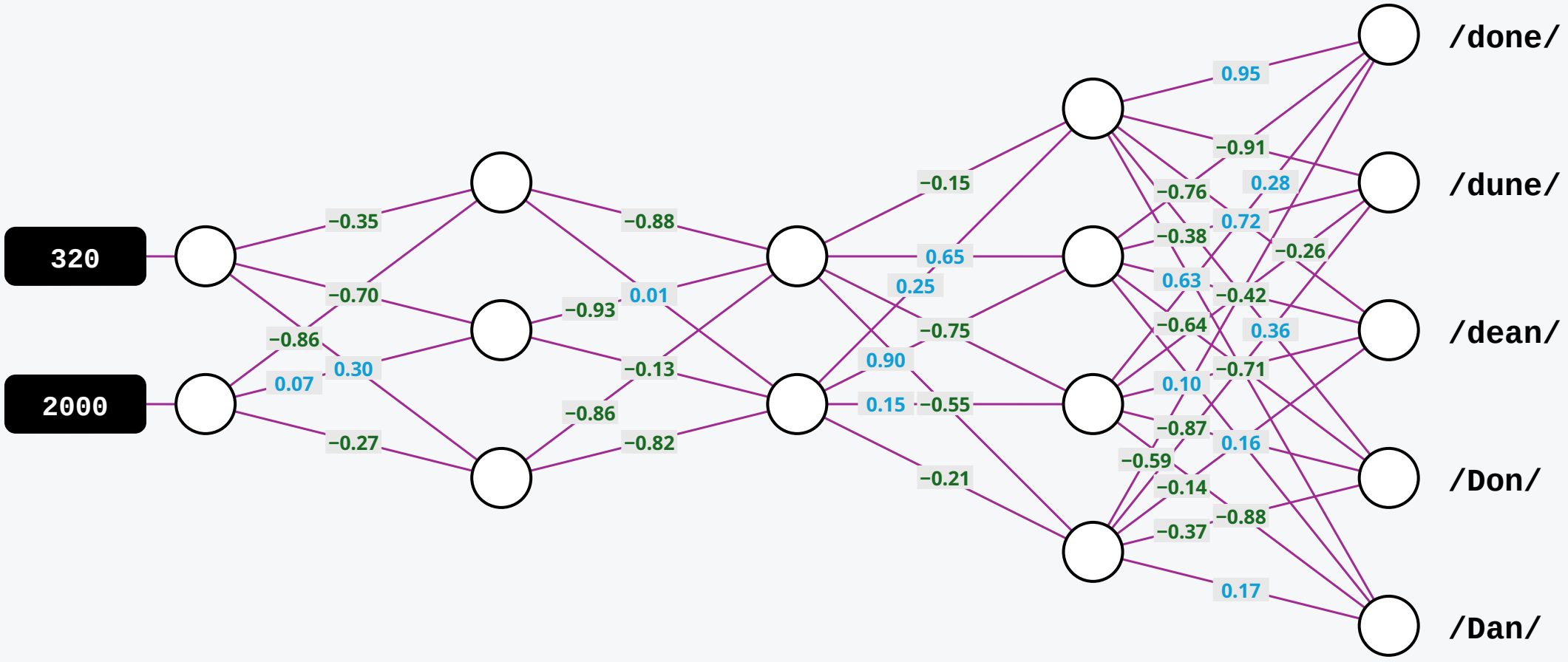
# A purely linear network

2 inputs → hidden layers of 3, 2 and 4 neurons → 5 outputs. No activation functions.



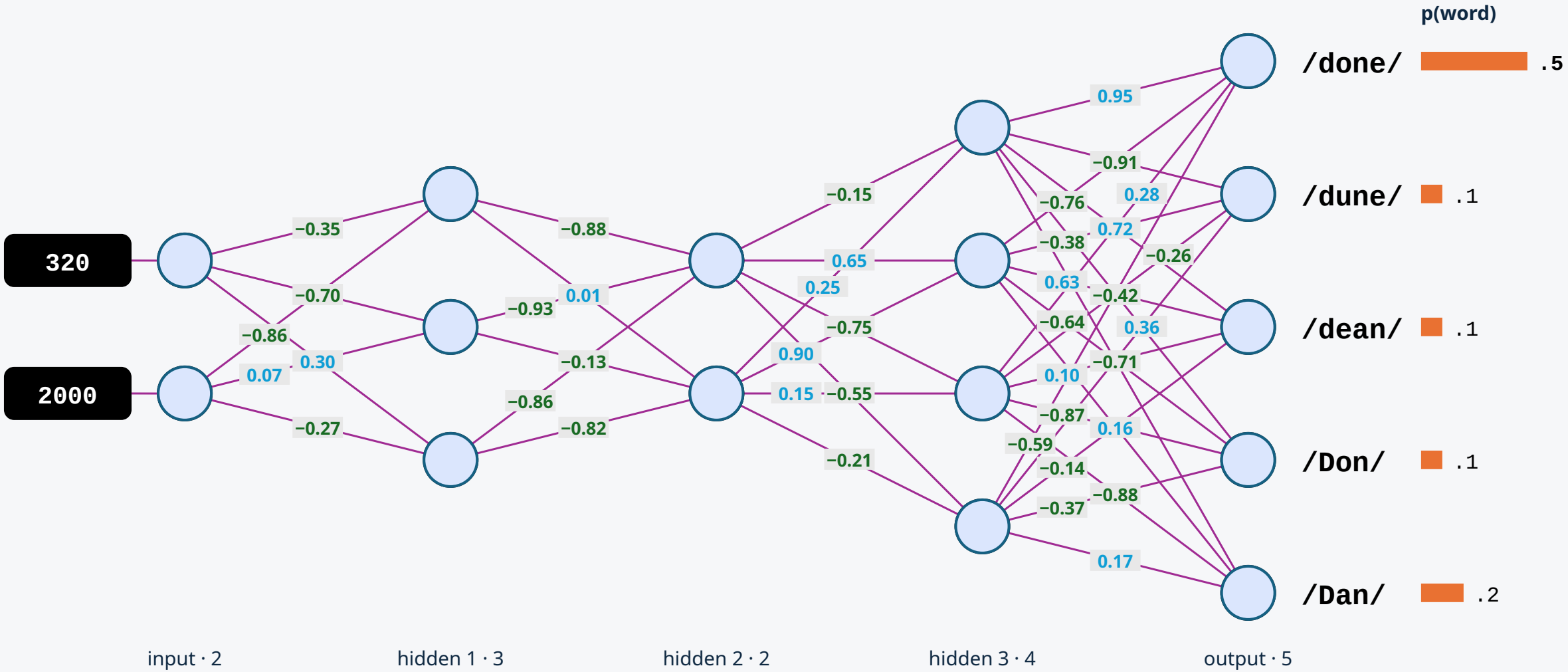
# An input, and random weights

Input (320, 2000). Every edge gets a random weight: **blue** positive, **orange** negative.

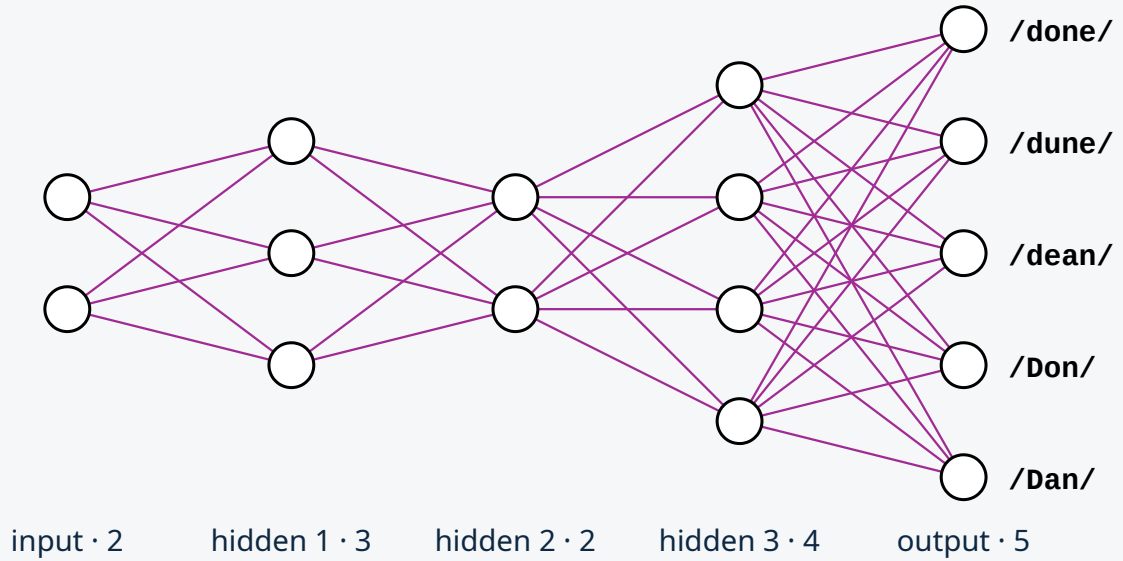


# Forward flow

The input flows forward, layer by layer. At the output: a distribution over the five words.



# The same network in PyTorch




```python
import torch
import torch.nn as nn

# purely linear: 2 → 3 → 2 → 4 → 5
net = nn.Sequential(
    nn.Linear(2, 3, bias=False),  # 2 in → 3 out
    nn.Linear(3, 2, bias=False),  # 3 in → 2 out
    nn.Linear(2, 4, bias=False),  # 2 in → 4 out
    nn.Linear(4, 5, bias=False),  # 4 in → 5 out
)
```

Declaration only, no training. `bias=False` keeps every layer purely linear, as in the picture.